DGMS Examination 2019 — Question 58 Detailed Solution

Junior Scientific Officer (Geology) | Structural Geology & Fold Geometry

QUESTION 58

Overfolds have one of the limbs essentially:

(a) Horizontal	**(b) Inverted** CORRECT ANSWER
(c) Vertical	(d) Parallel to the other

1. Executive Summary & Answer Verification

The correct option is **(b) Inverted** (also called *overturned*).

In structural geology (Billings, 1972; Davis & Reynolds, 1996; Haakon Fossen, 2016):

- An **Overfold (Overturned Fold)** is a fold in which the axial surface is inclined, and asymmetric tectonic contraction has pushed the fold past the vertical such that **both limbs dip in the same general direction**.
- Because one of the limbs has been rotated past the vertical ($>90^\circ$ from its original horizontal depositional position), the stratigraphic younging direction is completely reversed: older strata physically lie on top of younger strata. This limb is stratigraphically and geometrically **inverted**.
- Thus, an overfold is universally characterized by having one normal (upright) limb and one **inverted** limb.

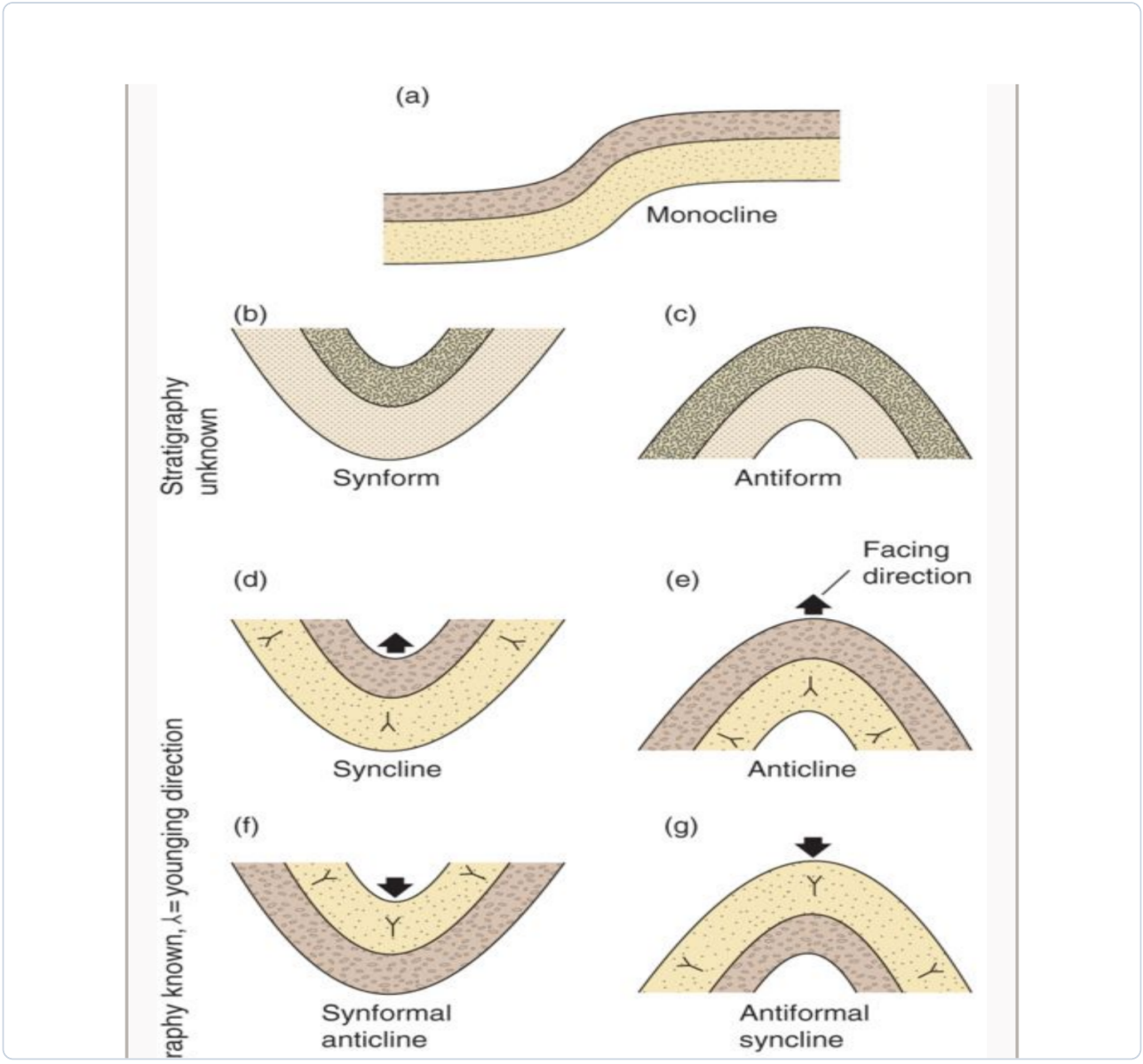


Figure 58.1: *Classification of fold geometries and limb younging relationships: Note that an overturned fold / overfold possesses an* ***inverted limb*** *where the younging direction (arrows) points downward toward the core. (Haakon Fossen, Structural Geology, 2nd ed., Chapter 12, Figure 12.8, p. 280).*

2. Comprehensive Structural Morphology & Classification

2.1 Geometric Definition of an Overfold

Folds are classified by the attitude of their axial surface and relative dip of their limbs:

1. **Symmetrical (Upright) Fold:** Axial plane is vertical. Both limbs dip in opposite directions at equal angles. Both limbs are upright/normal.
2. **Asymmetrical Fold:** Axial plane is inclined. Both limbs dip in opposite directions at unequal angles, but neither has been rotated past vertical. Both limbs remain upright.
3. **Overfold (Overturned Fold):** Axial plane is inclined. Compressive shear has rotated one limb past the vertical, causing both limbs to dip in the same direction (typically at unequal angles). One limb is normal, and the other is **inverted**.
4. **Recumbent Fold:** Axial plane is essentially horizontal (dip $< 10^\circ$). The fold lies on its side; one limb is upright, and the other inverted limb forms the lower thrust-plate boundary (nappe).

5. **Isoclinal Fold:** The interlimb angle is zero or near zero ($0^\circ\text{--}10^\circ$), meaning both limbs are **parallel to each other**. An isoclinal fold can be upright, overturned, or recumbent.

2.2 Criteria for Recognizing Inverted Limbs in the Field

When mapping overfolds, geologists determine which limb is inverted using primary sedimentary and structural geopetal (facing) indicators:

- **Cross-Bedding:** Foreset lamellae that are truncated at the top and asymptotic at the bottom indicate upright bedding; truncated lamellae at the bottom indicate an **inverted limb**.
- **Graded Bedding:** Coarse grains at the bottom grading upward into fine silt indicate upright strata; coarse grains at the top indicate an **inverted limb**.
- **Cleavage-Bedding Relationship:** In an upright limb, cleavage is steeper than bedding ($S_1 > S_0$); in an **inverted limb**, bedding is steeper than cleavage ($S_0 > S_1$).

3. Systematic Comparison of Fold Types by Limb & Axial Geometry

Fold Type	Attitude of Axial Plane	Limb Dip Direction	Limb Stratigraphic Status
Symmetrical Fold	Vertical (90°)	Opposite directions at equal angles	Both limbs upright (normal)
Asymmetrical Fold	Inclined	Opposite directions at unequal angles	Both limbs upright (normal)
Overfold (Overturned)	**Inclined**	**Same direction (unequal angles)**	**One limb upright, one limb INVERTED (Option b)**
Recumbent Fold	Horizontal or sub-horizontal ($< 10^\circ$)	Opposite/same sub-horizontal	One limb upright, one limb inverted (axial plane horizontal)
Isoclinal Fold	Any attitude	Limbs dip at identical angles	**Limbs are parallel to each other (Option d)**

4. Evaluation of All Options

- **(a) Horizontal — INCORRECT:** Describes a recumbent fold's axial plane, not an overfold limb.
- **(b) Inverted — CORRECT:** By structural definition, an overfold is pushed past vertical, making one of its limbs inverted (overturned).
- **(c) Vertical — INCORRECT:** A fold with a vertical limb is termed a *monocline* or an intermediate step, but overfolds have rotated past vertical.
- **(d) Parallel to the other — INCORRECT:** This defines an **isoclinal fold**, where limbs are mutually parallel.

Textbook Authority

Marland P. Billings, *Structural Geology* (Prentice Hall):
"An overturned fold, or overfold, is one in which the axial surface is inclined and both limbs dip in the same direction, usually at different angles. One limb is overturned or inverted."
P. K. Mukherjee, *Textbook of Geology*:

"In an overfold, the axial plane is inclined to such an extent that one of the limbs is turned completely upside down, that is, it is inverted. Consequently, both limbs dip in the same direction."